Core semantic digest: canonical coverage correction

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Scope and current evidence. The user requests correction of the four reproduced defects in Core 112a254. Prepare the exact patch against the clean audit base ae068b9, importing only the seven-file digest commit and correcting it. Do not import portable IDs, robotics fixtures or their corpus-golden changes. Core and provider fetched successfully at task start. The existing audit records 18 passing original tests and four failing identity requirements; reuse those exact negative controls.

Approval boundary. Core CORE.YAML requires a literal, slug-specific chat token. The user's request authorizes preparation; the token for protected application has not yet arrived. Proposed token: `APPROVED-CORE-EDIT: semantic-digest-canonical-coverage`. Prepare the patch, exact file inventory, invariants, tests and rollback first. Do not apply it to protected sources, including verification copies, before that token. After approval, quote the received token and add its commit footer.

Owner and affected files. The sole implementation owner remains Core ir::semantic_digest. Protected application paths: `CORE.YAML`; `hymeko_core/src/ir/hash_pass.rs`, `mod.rs` and `semantic_digest.rs` in that same directory; `hymeko_core/src/module_store/module_store.rs`. Existing test owner: `hymeko_core/tests/unified_ir/verification/mod.rs` and `semantic_digest.rs` in that same directory. Regenerate `SYMBOLS.md` using the installed provider tool. Proposal artifacts are this quartet, `reports/2026-09-06-semantic-digest-correction.md` and `reports/semantic-digest-correction` containing the review source, application patch, validation log and file hashes. Extend `.gitattributes` to preserve these evidence bytes. No new production dependency, IR or runner.

Canonical encoding. Keep public signatures. Build one private context from the existing path-index owner and an ArcRec-to-first-owner-slot table. Named targets use resolved UTF-8 path segments. Canonical anonymous targets use the owner's path and the first position in its ordered arc list. Ignore the anonymous declaration name: both the lowerer's SymId(0) and the test builder's generated name are placeholders. Preserve anonymous declaration annotations and parent identity.

Every ArcRec has its own domain-separated identity and full payload hash even without a HyperArc declaration. Edges fold those hashes in slot order, preserving duplicate references and sharing. Hash every concrete record table so reverse dispatch omissions cannot hide record payloads. Per-declaration output stays in DeclId order; only whole-model aggregation sorts the fixed-size hashes separately within each record-kind domain. Counts preserve multiplicity. Values use an iterative, order-preserving traversal with no depth truncation; preserve length, option, sign and exact floating-point bit encodings.

Canonical mode requires unique named paths and identifiable owner slots, with valid bindings for all anonymous declarations, at most one per ArcRec. Otherwise use the existing explicit, whole-model raw-ID fallback domain; hash all payloads and memberships there too. Promise reindexing invariance only in canonical mode. Document valid symbol and record indices as preconditions

and possible panics for violations; do not claim that bounded traversal is complete malformed-IR validation.

Version and preserved contracts. Schema stays 2; semantic algorithm version changes 1 to 2 because the encoded bytes change. Do not keep an old-algorithm runtime selector. V1 bytes/signatures, the deprecated Merkle body, Ir::decl_hash, CBOR schema, parser pin, distributed admission and the 19 existing Lean theorems remain unchanged. Rebuild any cached V2 digest; this is not a CBOR migration. Nagare's fetched f3660cc consumer test uses the current infallible signature; preserve that source compatibility and run its existing merkle_digest_soundness family after Core approval and tests.

Test and resource plan. Before approval, check patch applicability and Rust syntax/formatting only; these are not build or correctness results. After approval, apply in the existing audit checkout and promote the four recorded probes into the existing test owner. Add direct controls for anonymous placeholder/interner changes, shared identical arcs with [A,A,B] versus [A,B,B], orphan-record payload coverage, fallback idempotence, and depth-boundary values. Preserve the 18 incoming tests, updating only the explicit algorithm-version assertion. Check slot/sign/weight changes remain observable and V1 unchanged. Then run the existing complete Core gate, CBOR corpus, admission regressions, fmt and Clippy; do not re-freeze fixture hashes.

Reuse Amaterasu's existing SDK and task-owned Core verification object store, with an ephemeral clean source snapshot and existing cache after approval. 16 GiB memory, no extra swap, six CPUs, locked/offline tests. Record exact source, test-only deltas, command, elapsed time and RSS/cgroup peak. A substantive fix must pass those tests before any success claim or provider revision publication. Worst-case added storage is linear in declaration/record count plus an iterative value traversal stack; sorting costs O(R log R). No performance improvement claim. ETA: 15–25 minutes for preparation; after approval about 10–15 minutes for implementation and bounded Core/consumer verification, based on the prior 9.7-second focused rebuild and 84-second cold Core gate plus evidence handling.

Rollback and acceptance. Before approval, rollback is only removal/reversion of the proposal artifacts. After application use separate import/correction commits so the audit base and its tested consumer tags remain recoverable. Revert new downstream pins before reverting Core; recompute V2 caches for the selected algorithm version. Never move the previous audit tag. Acceptance requires all four original counterexamples green, new discriminating controls green, preserved V1/CBOR/admission gates and published exact provider provenance. No main merge or release is authorized.

